The Two-Player $3n \pm 1$ Game: A Binary Normal Form and a Well-Founded No-Draw Proof

Grisha Pochuev
n_854@mail.ru
https://github.com/Grisha-Pochuev/3n-plus-minus-1-game

Corrected manuscript, 10 August 2026

Audit status. This manuscript presents the repaired human proof after an external audit of an earlier version. The two invalid shortcuts identified by that audit remain withdrawn. Independent external re-audit of the repaired bridge is still pending, and no end-to-end Lean formalization is claimed.

Abstract

Consider the impartial two-player game on positive odd integers in which a move from $n > 1$ chooses $3n-1$ or $3n+1$ and removes all powers of two; reaching 1 wins immediately. Arbitrary play need not terminate, since $5 \to 7 \to 5$ is a legal cycle. The problem is whether optimal play nevertheless has finite remoteness from every start. We pass to a conjugated binary game with an expanding move A and a strictly contracting move B, introduce constant-tail coordinates, an embedded source map, and finite proof-height tokens for certified winning and losing positions. The global argument is a well-founded marked normalization: source decreases and certified token replacements form the outer rank, while a typed factor/obligation system is well-founded inside each fixed outer fibre. An external audit found that an earlier entry argument incorrectly identified a smaller canonical coefficient with a smaller coefficient source and also used a reverse-factor lemma outside its valid exponent range. We repair both issues. The key final step is a one-shot entry component that permits a returned inner source to be larger than the original source exactly once, during typed normalizer initialization, while forbidding any repeated unranked reset. A direct phase analysis of the remaining factor-free exponent-two base entry closes the last attachment case. The resulting human proof rules out all draw positions. Machine regressions and symbolic certificates are supporting checks with a strictly narrower scope than the human theorem.

Contents

1 The game and the theorem

The current state is a positive odd integer n. For $n > 1$ the player to move chooses one of

$$3n - 1, \qquad 3n + 1,$$

and divides the chosen value by 2 until it is odd. If the resulting odd value is 1, the player who made the move wins. Players alternate with perfect information.

For the player to move, a position is WIN if some move leads to a LOSS position, and is LOSS if every move leads to a WIN position. Because the game graph is infinite, these inductively generated classes need not a priori exhaust all positions. Every remaining position is called DRAW. This is the standard finite-remoteness interpretation of Conway's Beans-Don't-Talk game [1, 2]; the modern prize formulation is given by Althöfer [4, 3].

Theorem 1.1 (Main theorem)**.** *Every positive odd starting position has finite remoteness. Equivalently, the game has no* DRAW *positions, and optimal play reaches* 1 *after finitely many moves.*

The proof is not a termination proof for arbitrary play. For example,

$$5 \longrightarrow 7 \longrightarrow 5$$

is a legal two-ply cycle under suitable choices.

2 Binary conjugation

Write an odd original state as $n = 2m + 1$ and put

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil .$$

For a nonnegative integer x, let $R(x)$ be the integer obtained by deleting the maximal alternating suffix of the binary expansion of x. For example, $R(1001_2) = 10_2$ and $R(10101_2) = 0$.

Proposition 2.1 (First normal form)**.** *For $m > 0$, the two children in m-coordinates are exactly*

$$F(m), \qquad F(R(m)).$$

Proof. We have

$$3(2m+1)-1 = 2(3m+1), \qquad 3(2m+1)+1 = 2(3m+2).$$

Exactly one of $3m+1$ and $3m+2$ is odd. In the other expression, each additional factor of two removed corresponds to crossing one more alternating bit of the binary suffix of m. The division stops at the first repeated adjacent bit, which is precisely deletion of the maximal alternating suffix. The remaining odd value, returned to m-coordinates, is $F(R(m))$. $\square$

Every child in Proposition 2.1 lies in the image of F. Representing $F(q)$ by q gives the conjugated game

$$A(q) = F(q) = \left\lceil \frac{3q}{2} \right\rceil, \qquad B(q) = R(F(q)),$$

with terminal state $q = 0$.

Proposition 2.2 (Strict contraction)**.** *For every $q > 0$,*

$$B(q) < q.$$

Proof. Deleting a nonempty binary suffix gives

$$B(q) \le \left\lfloor \frac{F(q)}{2} \right\rfloor \le \left\lfloor \frac{3q+1}{4} \right\rfloor < q$$

for $q \ge 2$, and $B(1) = 0$ directly. $\square$

The difficulty is therefore concentrated in the expanding branch A and in the unbounded length of the alternating suffix deleted by B.

3 Constant-tail coordinates and source coordinates

For positive odd a, $r \ge 1$, and $e \in \{0, 1\}$ define

$$Q_r^e(a) = a2^r - e.$$

This is the canonical form of a positive binary word with a maximal constant suffix of r copies of the bit e.

Lemma 3.1 (Long-tail recurrence)**.** *For every $r \ge 3$,*

$$\boxed{A(Q_r^e(a)) = Q_{r-1}^e(3a), \qquad B(Q_r^e(a)) = Q_{r-2}^e(3a).}$$

Moreover the two boundary states share their contracting child:

$$\boxed{B(Q_1^e(a)) = B(Q_2^e(a)).}$$

Proof. For $r \ge 3$, the expanding child still ends in at least two equal bits, so its maximal alternating suffix consists only of the final bit. The boundary identity follows by comparing the binary words of $3a-e$ and $6a-e$; the additional appended bit extends the same alternating suffix and therefore deletes to the same prefix. $\square$

Define the embedded three-free coefficient

$$J(s) = 2A(s) + 1.$$

Every positive odd coefficient has a unique factorization

$$a = 3^k J(s), \qquad k \ge 0,$$

which defines its coefficient source s.

Lemma 3.2 (Embedded source transition)**.** *For $s > 0$, the two signed boundary transitions from the coefficient $J(s)$ select exactly the two ordinary children $A(s)$ and $B(s)$. More precisely, there is a phase*

$$\alpha(s) = 1 - \left(\left\lfloor \frac{s}{2} \right\rfloor \bmod 2 \right)$$

such that

$$\operatorname{odd}(3J(s) + 1 - 2\alpha(s)) = J(A(s))$$

with 2-adic valuation one, while the opposite phase selects $J(B(s))$ with valuation at least two.

The point of this embedding is that multiplication of a constant-tail coefficient by 3 preserves its source, while a signed exponent-one boundary reveals one ordinary A/B move on the source.

4 Finite outcome certificates and token rank

Every certified WIN or LOSS position has a finite canonical proof height:

$$h(0) = 0,$$

$$h(x) = 1 + \min\{h(y) : y \text{ is a LOSS child of } x\} \quad (x \text{ WIN}),$$

and

$$h(x) = 1 + \max\{h(y) : y \text{ is a child of } x\} \quad (x \text{ LOSS}).$$

A marked normalization configuration carries only finite states whose outcome and height comparison have actually been proved. These states are called proof tokens.

Lemma 4.1 (Multiset token rank)**.** *For a finite multiset M of proof heights define*

$$\Phi(M) = \mathop{\#}_{h \in M} \omega^h,$$

the natural ordinal sum of the monomials ω^h. Replacing one token of height H by any finite multiset of certified descendants of heights strictly below H strictly decreases Φ.

Proof. In Cantor normal form the coefficient of ω^H decreases by one, while all inserted terms have lower exponent. No collection of lower monomials can compensate for the loss at exponent H. □

This rank permits an exceptional local diamond to split one token into several certified descendants without losing well-foundedness.

5 Universal factor scan

Fix a three-free odd base $a = J(s)$ and a phase e. For $k \geq 0$ put

$$P_k = Q_2^e(3^k a), \qquad R_k = Q_1^e(3^k a),$$

$$C_k = B(P_k) = B(R_k), \qquad T_k = A(R_k).$$

Then

$$\operatorname{moves}(P_k) = \{R_{k+1}, C_k\}, \qquad \operatorname{moves}(R_k) = \{T_k, C_k\}.$$

Lemma 5.1 (Two-level DRAW horizon)**.** *If the adjacent frame* $\{P_k, R_k\}$ *contains a* DRAW*, then*

$$\boxed{\{C_k, T_k, C_{k+1}, T_{k+1}\} \text{ contains a DRAW.}}$$

Proof. The common child C_k cannot be LOSS, otherwise both frame members would be WIN. If C_k is DRAW we are done. If C_k is WIN and R_k is DRAW, then T_k is DRAW. The only remaining case has P_k DRAW and R_k non-draw; then R_{k+1} is DRAW, so one of its two children T_{k+1}, C_{k+1} is DRAW. □

Factor at either exposed level

$$3^{i+1}J(s) + 1 - 2e = 2^v J(t), \qquad v \geq 1.$$

Then

$$T_i = Q_v^{1-e}(J(t)).$$

If $v \geq 2$, the raw sibling is exactly

$$\boxed{C_i = Q_{v-1}^{1-e}(J(t)).}$$

If $v = 1$ and the positive raw sibling is DRAW, its coefficient source is strictly smaller than t. Thus every factorful exponent-one DRAW reaches a factor-free lift, a factor-free adjacent frame, or a genuine source descent within at most two consecutive factor levels.

A second reverse argument removes the unbounded factor exponent itself. If a factor frame does not pull back to a factor-free DRAW over the same source, then its only residual orientation is a marked gate

$$P_k \text{ LOSS}, \qquad R_k \text{ DRAW}.$$

The exact predecessor $Q_2^e(3^{k-1}J(s))$ is then either DRAW, in which case factors of three are removed at exponent two, or WIN, in which case its other child is an actual LOSS proof token. The nonexceptional side exposes a strictly lower WIN descendant; the exceptional side is an exact adjacent frame over that LOSS token. Hence no factor level creates an unmarked reset.

6 The typed fixed-fibre normalizer

The local source, factor, and proof-token lemmas form a finite collection of routing macros. Their full case proofs are recorded in Sections 91–135 of the proof supplement at the repository revision cited in Section 9. We use only the following assembled statement.

Proposition 6.1 (Typed normalizer)**.** *Suppose a marked entry has already been supplied with*

(i) a retained arithmetic source anchor,

(ii) a finite multiset of certified WIN/LOSS *proof tokens with proved provenance, and*

(iii) one of the typed obligation/factor control states of the supplement.

Then the equal-retained-data routing relation is well-founded.

The crucial scope restriction is that temporary routing cursors are not rank anchors. In particular a canonical continuation

$$O(x,e) \longrightarrow O(A(x), 1-e)$$

usually increases the displayed ordinary source x, but this $A(x)$ is only routing data. Sections 93–96 bound or rank the canonical A-streak and the subsequent B-selecting transfers. Sections 94–95 rank factor forks, and Sections 97–135 transport the finite tokens through every high-return and long-tail recurrence. Every operation that changes a retained source or token is supported by a strict comparison; pure routing preserves the retained projection and is well-founded inside the fixed fibre.

Lemma 6.2 (Well-founded fibre principle)**.** *Let a transition relation have an ordinal-valued projection π. If $C \to C'$ implies $\pi(C') \le \pi(C)$ and the relation restricted to every fibre $\pi^{-1}(\{\alpha\})$ is well-founded, then the full relation is well-founded.*

7 The external audit and the repaired entry

An external audit of an earlier version found two genuine defects.

First, the proof used

$$\text{smaller canonical coefficient} \Longrightarrow \text{smaller coefficient source},$$

which is false when powers of three are present. The explicit counterexample is

$$s = 1, \qquad J(1) = 5, \qquad a = 3J(1) = 15, \qquad e = 1.$$

The common boundary child is

$$R(3a - e) = R(44) = 22.$$

Its canonical coefficient is $11 < 15$, but $11 = J(3)$, so the source has increased from 1 to 3.

Second, the reverse-factor lemma was used at exponent one even though it is valid only from exponent two. Hence the family

$$Q_1^e(3^k J(s)), \qquad k > 0,$$

needed a separate entry proof. Section 7 gives that proof without reinstating either invalid shortcut.

7.1 One-shot initialization

The key observation is that there are two logically different operations: initializing the typed normalizer after the raw entry has been normalized, and resetting a normalizer that is already running. Only the second requires a preceding strict source/token decrease.

Refine the retained lexicographic rank by an entry bit

$$\eta \in \{1, 0\}, \qquad 1 > 0,$$

placed after the outer source and outer token multiset:

$$\boxed{(\text{outer source},\ \Phi(M),\ \eta,\ \text{inner source/token rank},\ \text{finite control rank}).}$$

Here $\eta = 1$ means that the current outer fibre has not yet initialized its typed inner normalizer, and $\eta = 0$ means that Proposition 6.1 is running.

The transition

$$\boxed{\eta : 1 \longrightarrow 0}$$

is strict independently of the numerical value assigned to the newly initialized inner source. Thus the first returned source may be larger than the original source. Within an unchanged outer source/token fibre, the bit can never return from 0 to 1. A later raw reinitialization is allowed only after an earlier outer source or outer proof-token component has strictly decreased, exactly as permitted by Lemma 6.2. Therefore the entry bit cannot hide a repeated source-growth cycle.

7.2 The factor-free exponent-two bootstrap

Let $x > 0$, $a = J(x)$, and put

$$P = Q_1^e(a), \qquad U = Q_2^e(a),$$

$$b = B(P) = B(U), \qquad F = A(U) = Q_1^e(3a), \qquad g = 1 - e.$$

Suppose U is DRAW at the globally least outer DRAW source x. The common child b has smaller canonical coefficient than $J(x)$, hence its coefficient source is below x; so b is not DRAW. A child of a DRAW cannot be LOSS, therefore

$$\boxed{b \text{ is WIN}, \qquad F \text{ is DRAW.}}$$

The exact first-factor identity is

$$\boxed{9a + 1 - 2e = 2^v J(b)}.$$

It follows that

$$A(F) = Q_v^g(J(b)).$$

We now split according to whether the input phase at x is B-selecting or A-selecting.

7.2.1 B-selecting input

Assume $e = 1 - \alpha(x)$. The first source-boundary numerator is divisible by 4. The consecutive numerator relation gives

$$9J(x) + 1 - 2e = 3(3J(x) + 1 - 2e) - 2(1 - 2e) \equiv 2 \pmod 4,$$

so

$$\boxed{v = 1.}$$

Writing $u = A(x)$ and using the parity that characterizes the B-selecting phase gives the exact common endpoint

$$\boxed{b = 3A(x) + 1.}$$

The two children of the DRAW state F are therefore

$$T = Q_1^g(J(b)), \qquad C = R(T).$$

The signed-prefix identity identifies the raw state C with the ordinary child of b selected by phase g:

$$\boxed{C \in \{A(b), B(b)\}.}$$

At least one of T, C is DRAW.

If T is DRAW, it is the direct exponent-one typed obligation at source b, and the one-shot bit initializes the normalizer.

If C is DRAW, the other ordinary child ℓ of the WIN state b is LOSS. For nonexceptional b, the ordinary side diamond exposes $B(A(b))$ as a WIN child of ℓ and as a child of the continuing DRAW, hence a strict proof-token/boundary descent. If b is exceptional and $C = A(b)$, the exceptional side formula gives an adjacent DRAW frame over the actual LOSS token ℓ.

The sole remaining exceptional orientation is $C = B(b)$ DRAW. Since $b = 3A(x) + 1$,

$$b \leq \frac{9x+5}{2}.$$

For exceptional b, the alternating suffix of $A(b)$ has length at least three, whence

$$C = B(b) \leq \frac{A(b)}{8} \leq \frac{3b+1}{16}.$$

If $\rho(C)$ is the coefficient source of C, then

$$\rho(C) < \frac{C}{6} \leq \frac{3b+1}{96} \leq \frac{27x+17}{192} < x.$$

Thus this last row is a genuine strict outer-source transition.

7.2.2 A-selecting input

Assume $e = \alpha(x)$. Lemma 3.2 gives

$$3J(x) + 1 - 2e = 2J(A(x)),$$

and the consecutive numerator identity implies

$$\boxed{v \geq 2.}$$

Retain

$$9J(x) + 1 - 2e = 2^v J(b).$$

If $v \geq 4$, then

$$16J(b) \leq 9J(x) + 1 \leq 27x + 19.$$

Since $J(b) \geq 3b + 1$,

$$48b + 16 \leq 27x + 19, \qquad \boxed{b \leq \frac{9x+1}{16} < x.}$$

Both children of F have factor-free source b, so this is a strict outer-source exit.

Only $v = 2, 3$ remain. If $v = 2$, F is a DRAW parent whose children are

$$Q_2^g(J(b)), \qquad Q_1^g(J(b)),$$

which is exactly the second form of the typed obligation $O(b, g)$.

If $v = 3$, the children are the valuation-three frame

$$Q_3^g(J(b)), \qquad Q_2^g(J(b)).$$

The constructor automatically supplies the congruence hypothesis needed by the first-factor hidden-parent lemma. Reducing

$$9J(x) + 1 - 2e = 8J(b)$$

modulo 3 and using $g = 1 - e$ yields

$$\boxed{J(b) \equiv 1 + g \pmod 3.}$$

Thus that hidden-parent lemma is used only in the typed rows for which its hypothesis is proved. The ensuing factor-frame normalization produces a strict source edge, a strict boundary/token edge, or the already typed obligation entry.

We have proved the following.

Proposition 7.1 (Factor-free base-entry attachment)**.** *Every factor-free exponent-two* DRAW *at a globally minimal outer source has an outcome-compatible finite continuation that either strictly lowers the retained outer source, strictly lowers a certified finite proof token, or consumes the one-shot entry bit and enters the typed normalizer.*

7.3 Arbitrary factorful exponent one

Return to

$$Q_1^e(3^k J(s)) \text{ DRAW}, \qquad k > 0.$$

The predecessor reduction described after Lemma 5.1 has two outputs. Either it returns to a factor-free DRAW over the same source, in which case the factor-free analysis terminates at exponent one or at the exponent-two bootstrap of Proposition 7.1; or it exposes a factor-free DRAW lift/frame together with an actual finite WIN/LOSS descendant of the predecessor token. In the latter case Lemma 4.1 supplies a strict token edge, after which Lemma 6.2 permits the inner arithmetic task to reset.

Consequently:

Proposition 7.2 (Repaired arbitrary exponent-one attachment)**.** *Every arbitrary factorful exponent-one* DRAW *has a finite outcome-compatible normalization that either strictly lowers the retained source, strictly lowers an already marked finite proof token, or consumes the one-shot entry bit and enters the typed normalizer. No comparison $t < s$ is required for the temporary returned inner source.*

8 Proof of the main theorem

Proof of Theorem 1.1. Assume that a DRAW exists and choose the least coefficient source s of a DRAW. The source-zero case is handled by the explicit zero-source constant-tail normalization in the supplement. Assume $s > 0$.

The constant-tail/source analysis and the factor scan reduce the continuing actual DRAW to a marked macro-configuration. By Proposition 7.2, every macrostep has one of three effects:

(1) the retained outer source strictly decreases;

(2) at fixed outer source, the finite proof-token multiset strictly decreases under Φ;

(3) with both outer components fixed, the one-shot bit changes from $\eta = 1$ to $\eta = 0$, after which the route lies in the typed fixed-fibre normalizer of Proposition 6.1.

The first effect contradicts minimality when it occurs at the initial outer source and, more generally, is a strict first lexicographic component. The second is well-founded by Lemma 4.1. The third is strict at the entry bit and cannot reset back to 1 while the preceding components are unchanged. Once $\eta = 0$, Proposition 6.1 gives well-foundedness of the equal-retained-data fibre. Lemma 6.2 therefore makes the complete marked macro relation well-founded.

On the other hand, every DRAW position has no LOSS child and has at least one DRAW child. Hence from any marked DRAW configuration one may choose an outcome-compatible continuing DRAW and repeat the finite normalization macrostep forever. This would produce an infinite path in the well-founded marked relation, a contradiction.

Therefore the conjugated game has no DRAW positions. Conjugacy with the original odd-integer game proves that every positive odd starting state has finite remoteness and optimal play reaches 1. □

9 Verification boundary and reproducibility

The mathematical claim above is a human proof. The computational artifacts have deliberately narrower roles.

- The repository's standard-library Python tests check exact arithmetic identities, finite outcome certificates, and finite regression ranges.
- The symbolic global-routing JSON checks the declared typed control/rank assembly. It does not encode the new one-shot η bridge and therefore remains a conditional machine check rather than an end-to-end proof.
- The Lean project kernel-checks general metatheory and selected definitions, but does not kernel-check the complete concrete no-DRAW theorem.

The corrected proof package used for this manuscript is the branch `repair/althoefer-audit-gap` of `https://github.com/Grisha-Pochuev/3n-plus-minus-1-game`, with the closure addendum `docs/althoefer-audit-closure.md`. At the time this manuscript was prepared, the corresponding draft pull request was PR #1. The earlier archived preprint is `https://doi.org/10.5281/zenodo.21844684`. That archived version predates the external audit and should not be used in place of this corrected manuscript.

The companion file `verify_closure.py` supplied with this manuscript is a standalone arithmetic regression for the new repair identities. It has no third-party dependencies and explicitly labels itself as supporting verification, not as a proof.

10 Audit checklist for the repaired bridge

An independent reviewer should try to falsify the following points first:

1. no step identifies a smaller canonical coefficient with a smaller coefficient source;
2. the reverse-factor lemma is never used at exponent one;
3. the B-selecting exponent-two base row really has factor valuation one;
4. the exceptional raw B-child estimate gives a source strictly below the old source;
5. the A-selecting row with $v \geq 4$ really has $b < x$;
6. the $v = 3$ constructor really satisfies the additional hidden-parent congruence before that lemma is invoked;
7. the entry bit is placed after the outer source/token components and never changes $0 \to 1$ inside an unchanged outer fibre;
8. every later reset is preceded by a genuine strict retained source or proof-token decrease;

9. each macro follows an actual outcome-compatible continuing DRAW, while finite proof tokens are kept separate from temporary arithmetic cursors.

Acknowledgment of the audit

The repaired version was prompted by questions from Ingo Althöfer concerning the former global entry argument. Those questions led to the explicit counterexample to the old source-descent inference and to the separate proof of the exponent-one entry used here.

References

[1] R. K. Guy, *John Isbell's Game of Beanstalk and John Conway's Game of Beans-Don't-Talk*, Mathematics Magazine 59(5) (1986), 259–269. DOI: `https://doi.org/10.1080/0025570X.1986.11977258`.

[2] R. K. Guy, *Unsolved Problems in Combinatorial Games*, Problem 42, in *Games of No Chance*, MSRI Publications 29, Cambridge University Press, 1996.

[3] I. Althöfer, M. Hartisch, T. Zipproth, *Analysis of a Collatz Game and Other Variants of the* $3n + 1$ *Problem*, Advances in Computer Games (2024), 123–132. DOI: `https://doi.org/10.1007/978-3-031-54968-7_11`.

[4] I. Althöfer, *Collatz Problems: The Two-Player* $3n{+}{-}1$ *Game*, `https://althofer.de/collatz-prizes.html`, accessed August 2026.

[5] G. Pochuev, *Optimal* $3n \pm 1$ *Game: Proof and Verification Repository*, `https://github.com/Grisha-Pochuev/3n-plus-minus-1-game`, 2026.

[6] G. Pochuev, *Optimal Play in the Two-Player* $3n \pm 1$ *Game*, Zenodo record `https://doi.org/10.5281/zenodo.21844684`, 2026. The record predates the external audit discussed in this corrected manuscript.